LAB 06: PYTHON MATPLOTLIB

Introduction to Data Visualization

Course: Python / Data Structures Lab	Lab Number: 06
Student Name: _____	Date: _____

1. Introduction

Matplotlib is a Python library used to create graphs and charts. It helps us understand data visually instead of looking only at numbers. It is commonly used in data science, machine learning, and artificial intelligence.

2. Lab Objectives

1. Understand the basics of Matplotlib and its importance in data visualization.
2. Create different plots, including line plots, bar charts, scatter plots, histograms, and pie charts.
3. Customize plots using titles, labels, colors, legends, and grids.

3. Installing and Importing Matplotlib

Run the following command in Jupyter Notebook only if Matplotlib is not already installed:

```
%pip install matplotlib
```

Import the library:

```
import matplotlib.pyplot as plt
```

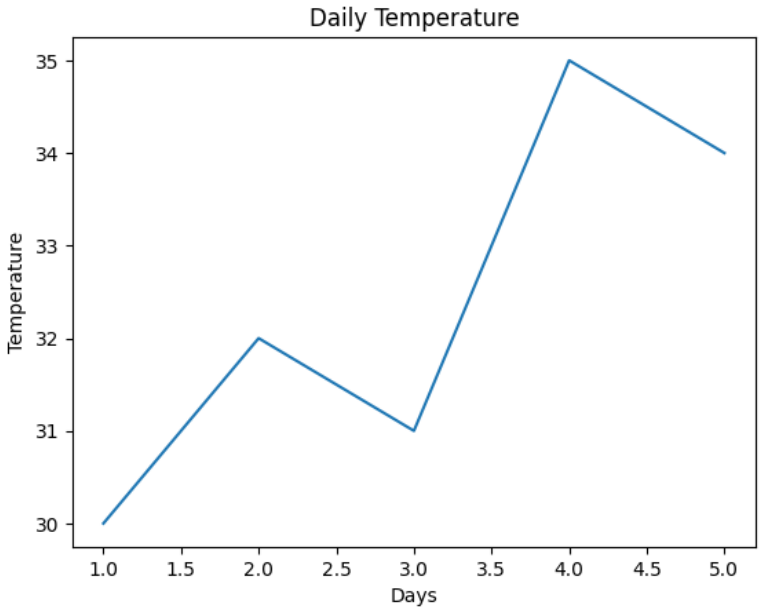
4. Line Plot

A line plot shows changes in data over time.

```
days = [1, 2, 3, 4, 5]
temperature = [30, 32, 31, 35, 34]

plt.plot(days, temperature)
plt.title("Daily Temperature")
plt.xlabel("Days")
plt.ylabel("Temperature")
plt.show()
```

Plot Output:



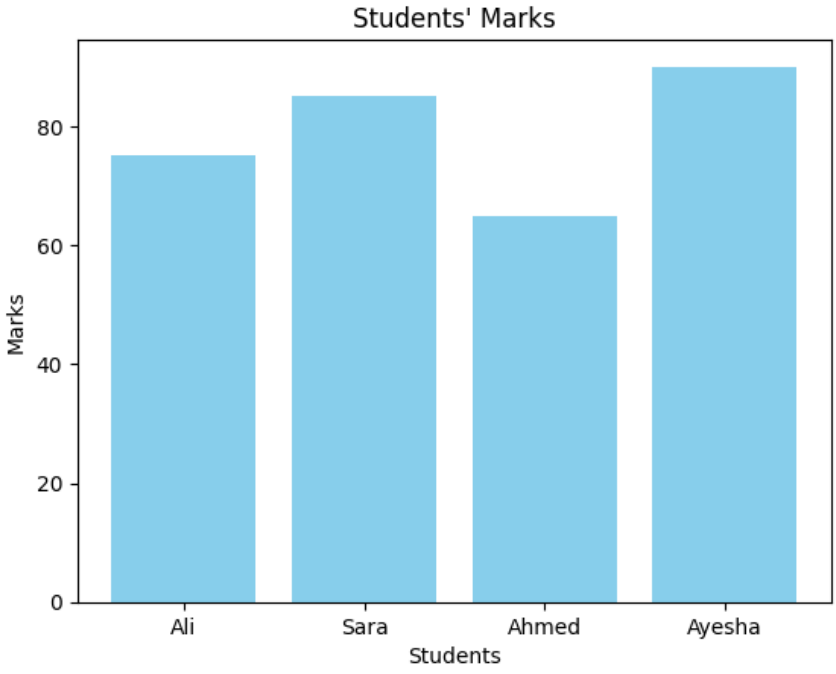
5. Bar Chart

A bar chart compares values between different categories.

```
students = ["Ali", "Sara", "Ahmed", "Ayesha"]
marks = [75, 85, 65, 90]

plt.bar(students, marks, color="skyblue")
plt.title("Students' Marks")
plt.xlabel("Students")
plt.ylabel("Marks")
plt.show()
```

Plot Output:



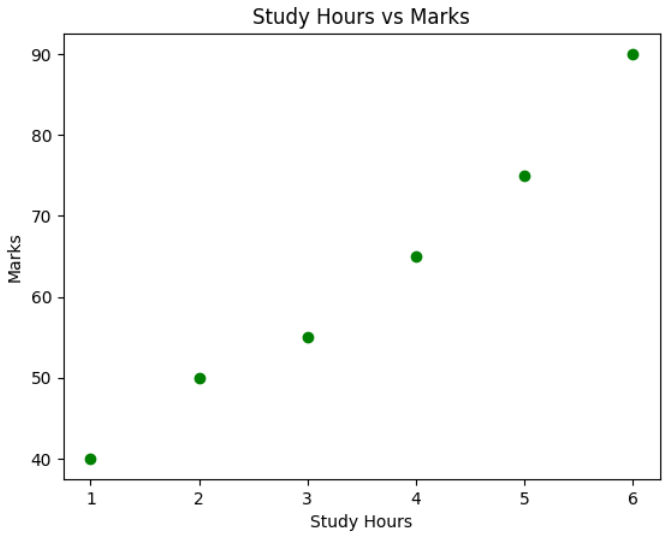
6. Scatter Plot

A scatter plot shows the relationship between two variables.

```
hours = [1, 2, 3, 4, 5, 6]
marks = [40, 50, 55, 65, 75, 90]

plt.scatter(hours, marks, color="green")
plt.title("Study Hours vs Marks")
plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.show()
```

Plot Output:



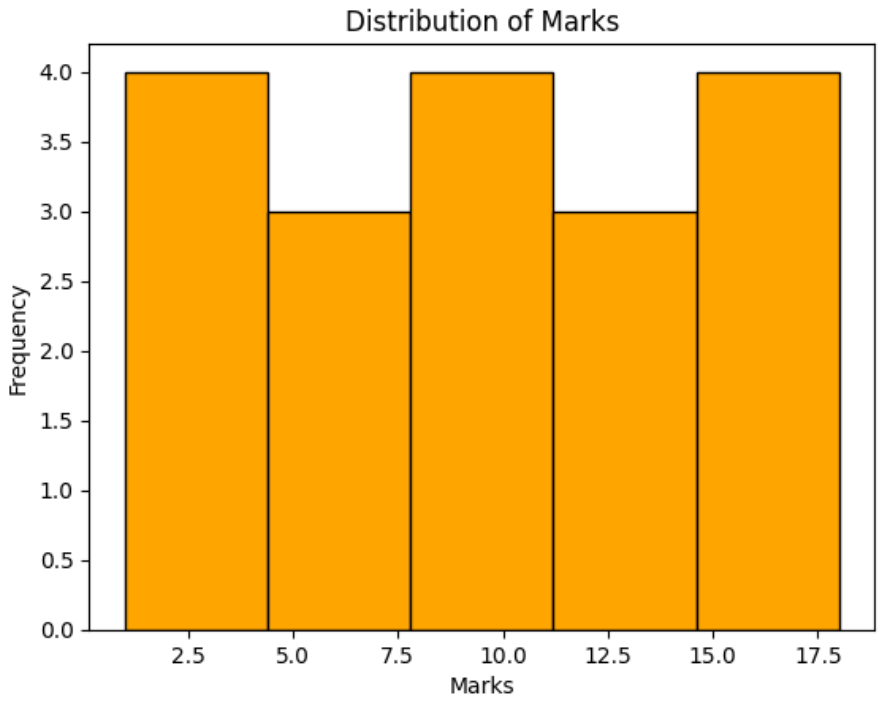
7. Histogram

A histogram shows the distribution of numerical data.

```
marks = [1,2,3,4,5,6,7,8,9,10,12,13,14,15,16,17,18]

plt.hist(marks, bins=5, color="orange", edgecolor="black")
plt.title("Distribution of Marks")
plt.xlabel("Marks")
plt.ylabel("Frequency")
plt.show()
```

Plot Output:



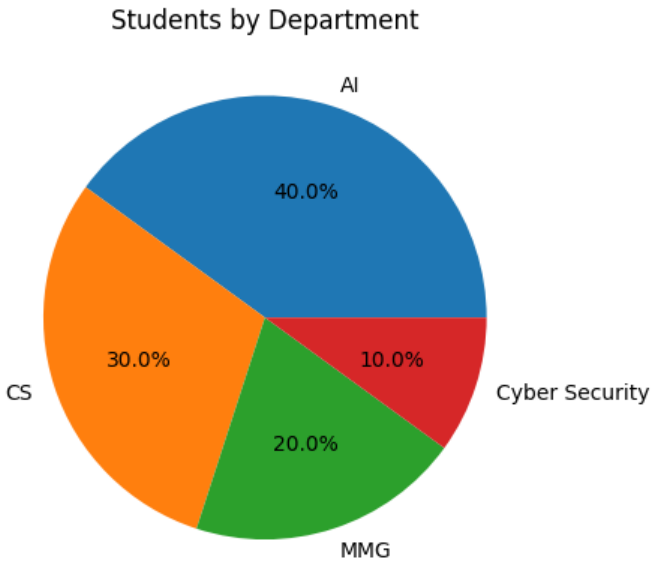
8. Pie Chart

A pie chart shows the proportion of different categories.

```
departments = ["AI", "CS", "MMG", "Cyber Security"]
students = [40, 30, 20, 10]

plt.pie(students, labels=departments, autopct="%1.1f%%")
plt.title("Students by Department")
plt.show()
```

Plot Output:



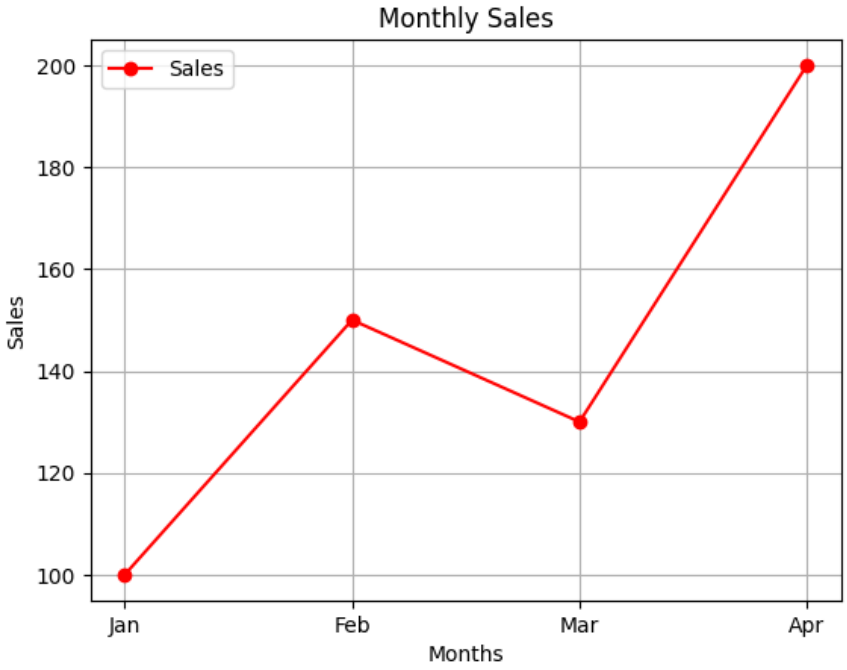
9. Adding Colors, Labels, and Grid

We can customize graphs to make them easier to understand.

```
months = ["Jan", "Feb", "Mar", "Apr"]
sales = [100, 150, 130, 200]

plt.plot(months, sales, color="red", marker="o", label="Sales")
plt.title("Monthly Sales")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.legend()
plt.grid(True)
plt.show()
```

Plot Output:



10. Multiple Plots Using Subplots

Subplots allow us to display multiple graphs in one figure.

```
students = ["Ali", "Sara", "Ahmed"]
marks = [70, 85, 90]

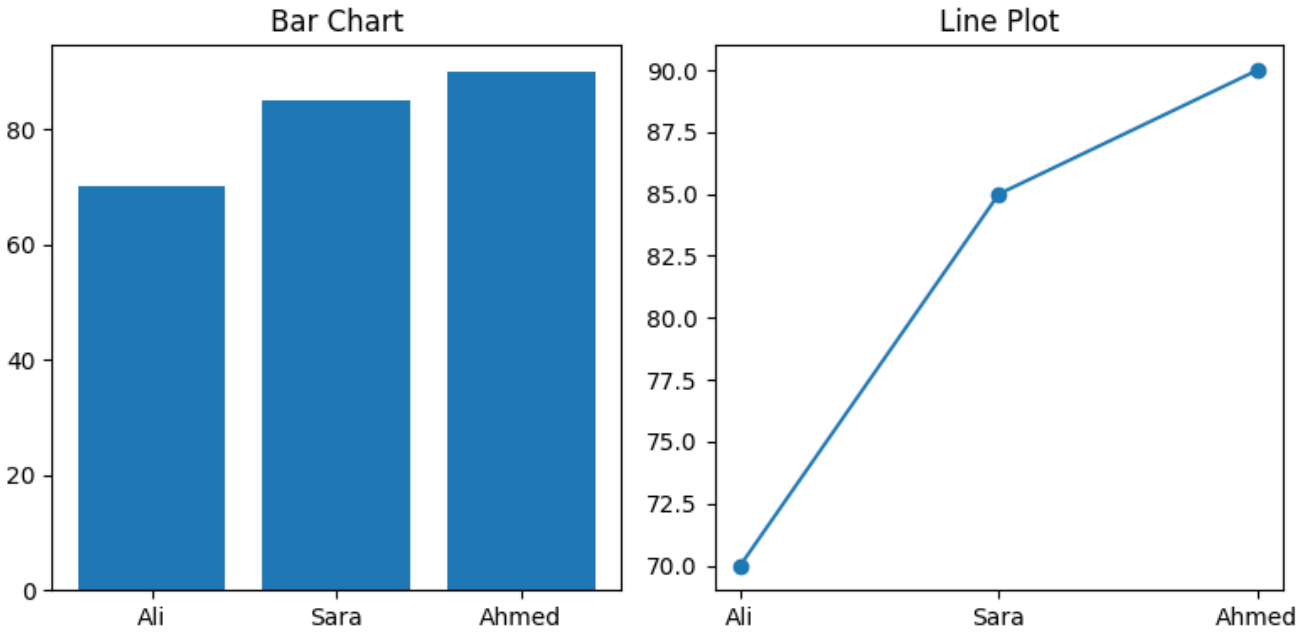
fig, axes = plt.subplots(1, 2, figsize=(8, 4))

axes[0].bar(students, marks)
axes[0].set_title("Bar Chart")

axes[1].plot(students, marks, marker="o")
axes[1].set_title("Line Plot")

plt.tight_layout()
plt.show()
```

Plot Output:



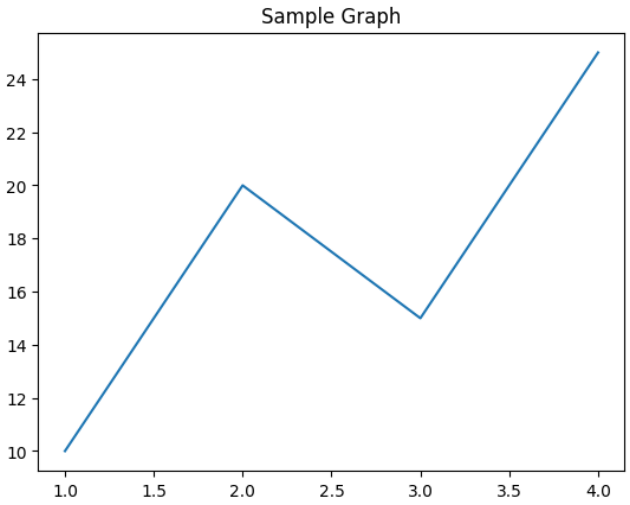
11. Saving a Graph

We can save a graph as an image file.

```
plt.plot([1, 2, 3, 4], [10, 20, 15, 25])
plt.title("Sample Graph")

plt.savefig("my_graph.png")
plt.show()
```

Plot Output:



12. Common Matplotlib Functions

Function	Purpose
plt.plot()	Creates a line plot
plt.bar()	Creates a bar chart
plt.scatter()	Creates a scatter plot
plt.hist()	Creates a histogram
plt.pie()	Creates a pie chart
plt.title()	Adds a title
plt.xlabel()	Labels the x-axis
plt.ylabel()	Labels the y-axis
plt.legend()	Displays a legend
plt.grid()	Displays grid lines
plt.show()	Displays the graph
plt.savefig()	Saves the graph

13. Student Tasks

A.

- Create a line plot showing the temperature for seven days.
- Create a bar chart showing the marks of five students.
- Create a scatter plot showing the relationship between study hours and marks.
- Create a pie chart showing students in four departments.
- Create a histogram using at least 20 examination marks.

B.

- Create a line plot with a title, axis labels, colors, markers, and grid.
- Create two subplots: one bar chart and one line plot.
- Create a histogram and change the number of bins.
- Create a pie chart displaying percentages.
- Save a customized graph as a PNG image.

14. Conclusion

In this lab, students learned the basics of Matplotlib and practiced creating different types of graphs. They also learned how to customize plots, display multiple graphs, and save visualizations as images. These skills are useful for understanding and presenting data in Python.